

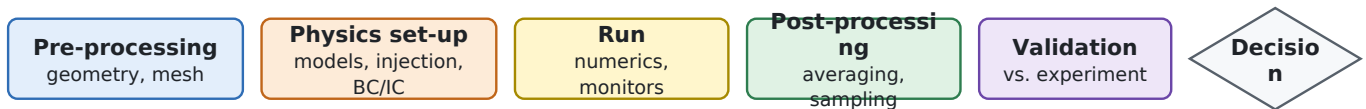
Bubbly flow in a vertical tube

Euler-Euler CFD workflow with OpenFOAM v2606, validated against LDA and bubble imaging

Item	Value / assumption
Tube	Vertical, inner diameter $D = 30$ mm, height $H = 350$ mm, open at the top
Liquid	Water, static initial column $H_L = 300$ mm, 50 mm air head space above it
Gas	Air injected at the bottom, $Q = 0.1$ - 0.5 L/min (superficial $j_g = 2.4$ - 11.8 mm/s)
Measurements	LDA (liquid and gas velocities) on horizontal planes at chosen heights z_i ; imaging for bubble size, shape and rise velocity
Hard constraints	Euler-Euler solver; only models that ship with OpenFOAM v2606 (no user code)
Temperature	Isothermal, about 20 °C (the solver still solves energy, so T is fixed via ICs and BCs)

How to use this document

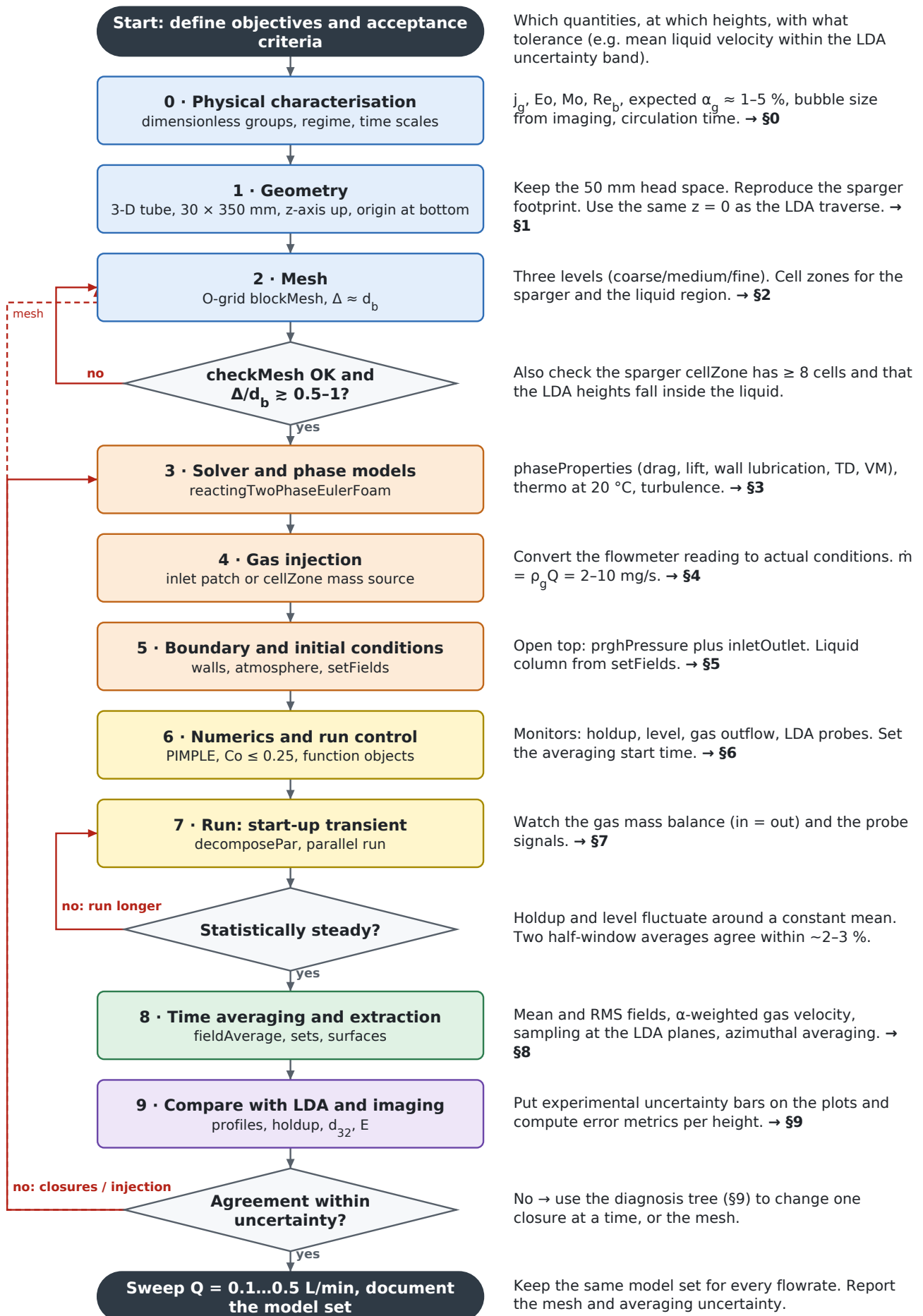
Page 2 shows the whole workflow as one flowchart. Each block in it is a stage (0-9), and each stage has its own section with a detailed sub-flowchart, the OpenFOAM v2606 keywords to use, and what to check before moving on. Model names in monospace are the exact run-time type names I checked in the v2606 sources and tutorials. Appendix A has a template case that was generated from these recommendations and checked with blockMesh, checkMesh, topoSet, setFields and a short reactingTwoPhaseEulerFoam run.



Summary of the recommendations

- **Solver:** reactingTwoPhaseEulerFoam (two-fluid, one shared pressure). It is the v2606 Euler-Euler solver with the full set of bubbly-flow closures: drag with aspect-ratio and swarm corrections, wall-damped lift, wall lubrication, turbulent dispersion, virtual mass, bubble-induced-turbulence RAS models, and an optional population balance.
- **Domain:** full 3-D tube including the 50 mm air space, so the free surface and level swell come out of the simulation. Use an O-grid mesh with cell size $\Delta \approx 1.5$ - 3 mm (of the order of the bubble diameter). Do not refine far below d_b .
- **Bubbles:** 2-6 mm air bubbles in water are in the ellipsoidal/wobbling regime ($Eo \approx 0.5$ - 5 , $Re_b \approx 400$ - 1500). Take the mean diameter from imaging, not from a correlation. The Tomiyama lift coefficient changes sign at $d_b \approx 5.8$ mm, so the measured size decides whether gas peaks at the wall or on the axis.
- **Turbulence:** the flow is turbulent (bubble $Re \sim 10^3$, recirculation $Re_D \sim 10^3$ - 10^4), so use RAS. Baseline: LaheyKEpsilon for water with continuousGasKEpsilon for air. Alternatives: mixtureKEpsilon and kOmegaSSTsato.
- **Injection:** for a porous plate or multi-hole sparger, use an inlet patch ($\alpha_{air} = 1$, flowRateInletVelocity). For a single needle, use a mass source on a small cellZone (scalarSemiImplicitSource on thermo:rho.air, plus the matching e.air source).
- **Comparison:** time-average after the start-up transient, sample two orthogonal diameters and full planes at every LDA height, average azimuthally, and compare the liquid mean and RMS, the α -weighted gas velocity, the slip, the void profile, the global holdup and level swell, and d_{32} and aspect ratio.

Master flowchart



§0 · Physical characterisation (before OpenFOAM)

These estimates tell you which closures matter, what mesh you can afford and how long to average. Properties: water at 20 °C ($\rho_l = 998 \text{ kg/m}^3$, $\mu_l = 1.0 \text{ mPa}\cdot\text{s}$, $\sigma = 0.072 \text{ N/m}$), air $\rho_g = 1.2 \text{ kg/m}^3$. Morton number $Mo = g\mu_l^4\Delta\rho/(\rho_l^2\sigma^3) = 2.7\text{e-}11$. Tube cross-section $A = 7.07 \text{ cm}^2$.

Gas flowrates

Q [L/min]	Q [m³/s]	$j_g = Q/A$ [mm/s]	$\dot{m} = \rho_g Q$ [kg/s]	α_g est. (drift flux) [%]	level swell Δh [mm]	bubble freq. @ 3 mm [1/s]
0.1	1.667e-06	2.36	2.00e-06	0.9	2.8	118
0.2	3.333e-06	4.72	4.00e-06	1.8	5.6	236
0.3	5.000e-06	7.07	6.00e-06	2.7	8.5	354
0.4	6.667e-06	9.43	8.00e-06	3.6	11.3	472
0.5	8.333e-06	11.79	1.00e-05	4.5	14.0	589

Holdup from the drift-flux model $\alpha_g = j_g/(C_0 j + V_{gj})$ with $C_0 = 1.2$ and $V_{gj} = 1.53(\sigma g \Delta\rho/\rho_l^2)^{1/4} = 0.25 \text{ m/s}$. There is no net liquid flow, so $j = j_g$. Level swell $\Delta h = H_L \alpha_g/(1-\alpha_g)$. These are order-of-magnitude checks for the simulation, not validation data. The level swell stays well inside the 50 mm head space.

Bubbles (take the actual size from imaging)

d_b [mm]	E_o [-]	$E = b/a$ (Wellek)	U_T [m/s] slightly contam.	U_T [m/s] fully contam.	Re_b [-]	C_L Tomiyama	d_b/D	D/d_b cells @ $\Delta=d_b$
2	0.54	0.91	0.286	0.208	570	+0.29	0.07	15
3	1.22	0.84	0.251	0.251	749	+0.29	0.10	10
4	2.17	0.77	0.236	0.236	940	+0.29	0.13	8
5	3.40	0.71	0.231	0.231	1150	+0.18	0.17	6
6	4.89	0.65	0.231	0.231	1382	-0.04	0.20	5

U_T from the Tomiyama (1998) drag correlation for a single bubble in stagnant water. Tap water is usually 'slightly contaminated'. C_L is the Tomiyama et al. (2002) lift coefficient evaluated with the slightly-contaminated U_T .

What these numbers mean for the set-up

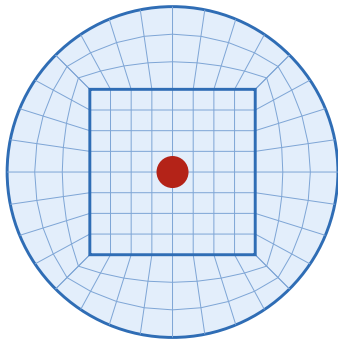
- Regime:** with $j_g \leq 12 \text{ mm/s}$ and $\alpha_g \leq 5\%$, the flow is homogeneous bubbly with no coalescence-driven transition. A fixed bubble diameter (constant or isothermal) is a sound baseline. Hydrostatic expansion over 0.3 m changes d_b by only $\sim 1\%$.
- Bubble shape:** $E_o \approx 0.5\text{--}5$ means ellipsoidal, wobbling bubbles. Drag depends on shape and contamination, so choose IshiiZuber, TomiyamaCorrelated or TomiyamaAnalytic (with an aspect-ratio model), not a sphere law. Imaging gives $E = b/a$ directly, so compare it with Wellek/VakhrushevEfremov.
- Lift sign:** the Tomiyama C_L is positive below $d_b \approx 5.8 \text{ mm}$, which pushes bubbles toward the wall (wall-peaked void). It is negative above that size (core-peaked). **The measured d_b and E are the most important experimental inputs to the model.**
- Confinement:** $d_b/D \approx 0.07\text{--}0.2$. Euler-Euler averaging assumes many bubbles per cell, so the tube can hold only $\sim 10\text{--}15$ cells across. This limits mesh refinement (§2) and makes the near-wall region the least reliable part of the comparison.
- Plume or column:** a single needle produces a meandering bubble chain or plume, not a uniform dispersion. Euler-Euler RANS predicts only the time-averaged spreading, which depends heavily on turbulent dispersion and on the injection model. Compare time-averaged quantities, and expect weaker agreement in the lowest plane.
- Time scales:** bubble transit time $H_L/U_T \approx 1.2\text{--}1.5 \text{ s}$. Liquid circulation (up in the core, down at the wall) is $\sim 0.1 \text{ m/s}$, so one loop takes $\sim 2H_L/U_l \approx 5\text{--}6 \text{ s}$. Plan $\sim 10\text{--}20 \text{ s}$ of start-up and $30\text{--}60 \text{ s}$ of averaging (§7).
- Turbulence:** $Re_b \sim 500\text{--}1500$ gives turbulent wakes (bubble-induced turbulence, BIT). The recirculation $Re_D = U_l D/\nu \approx 3000$ is transitional to turbulent. A laminar Euler-Euler model is not physically justified here and is useful only as a debugging baseline.

§1 · Geometry

- **Full 3-D cylinder.** A 2-D axisymmetric wedge cannot represent the 3-D meandering plume, and forces the gas onto the axis. Use it only for quick model screening if the sparger is a centred, uniform porous disc.
- **Include the air head space** ($z = 300\text{--}350$ mm). The free surface then develops naturally, and the level swell becomes a measurable output. The top patch is the open atmosphere.
- **Coordinates:** z -axis up, origin at the centre of the tube bottom, $g = (0\ 0\ -9.81)$. Define the LDA heights z_i in the same frame as the rig: from the bottom plate or from the sparger tip, and write down which.
- **Sparger:** record its type (needle, orifice plate, porous disc), orifice diameter, number of holes and position (centred or eccentric). This sets the injection model in §4. A needle that protrudes into the tube can be ignored as a solid if its $OD \ll D$, but then inject at its tip height.
- **Wall material and optical box:** not modelled, but note them. The curved tube wall refracts the LDA beams, so the rig should use a square index-matching water jacket or apply a refraction correction to the measurement-volume position. Otherwise near-wall points are misplaced.

§2 · Mesh

Build the mesh with `blockMesh` as an O-grid (butterfly): a square core block plus four outer blocks with arc edges. This avoids the degenerate cells a wedge-type mesh has on the axis, and the mesh is fully hexahedral and orthogonal near the wall. Appendix A has a `blockMeshDict` that uses `#eval` for the vertex coordinates.



cross-section: core block $N_c \times N_c$,
4 outer blocks $N_c \times N_r$ with arc edges

● sparger cellZone (`cylinderToCell`)

Keep the core-block cells and the arc cells
about the same size (core half-side $\approx R/2$).

Axial: uniform $\Delta z \approx \Delta$ (aspect ratio $\leq \sim 3$).
Optional: grade Δz finer at the free surface
($z \approx 0.28\text{--}0.33$ m) to sharpen the level.

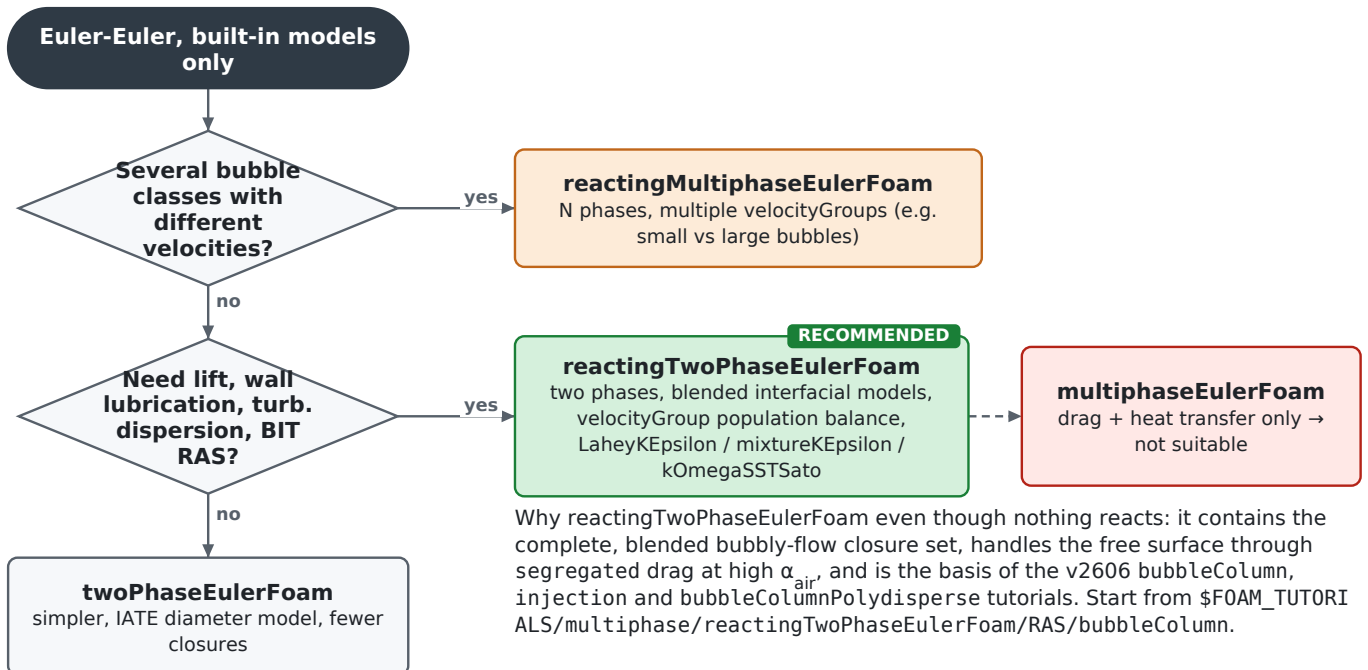
Level	$N_c / N_r / N_z$	cells	Δ (core / arc / Δz)	Δ/d_b (3 mm)	Use
Coarse	8 / 3 / 117	≈ 26 k	1.9 / 2.9 / 3.0 mm	$\approx 0.6\text{--}1.0$	model screening, quick parameter sweeps
Medium	12 / 4 / 175	≈ 59 k	1.25 / 2.0 / 2.0 mm	$\approx 0.4\text{--}0.7$	baseline production mesh (checked: <code>checkMesh</code> OK, max non-orth 27°)
Fine	16 / 5 / 230	≈ 128 k	0.9 / 1.5 / 1.5 mm	$\approx 0.3\text{--}0.5$	sensitivity only. Below $\Delta \approx d_b/2$ the averaging assumption breaks down

Mesh independence in Euler-Euler bubbly flow is not classical grid convergence. The closures were calibrated for cells larger than the bubbles. Refining further moves the model out of its validity range, and results may drift instead of converging. Judge mesh adequacy by the plateau of integral quantities (global holdup, level swell, centreline liquid velocity at each LDA plane) between the coarse and medium meshes. Report the medium-to-fine difference as a modelling uncertainty.

- **Cell zones** (`topoSet`): sparger (`cylinderToCell` around the orifice, $\geq 2 \times 2 \times 2$ cells) and `liquidRegion` (a box from just above the sparger to just below the free surface, used for the holdup monitor).
- **Walls:** the first cell centre sits at $\approx 0.5\text{--}1$ mm from the wall. With a near-wall liquid velocity of $\sim 0.05\text{--}0.1$ m/s, $y^+ \approx 3\text{--}10$, which is in the buffer layer. Use wall functions (`nutkWallFunction`) with the $k\text{--}\epsilon$ family and check `yPlus`. With `kOmegaSSTsato` use `omegaWallFunction`, which blends between the viscous and log layers.
- **Always run** `checkMesh`. Target: non-orthogonality $< 40^\circ$, skewness < 2 , aspect ratio < 5 .

§3 · Solver and phase models

v2606 ships four Euler-Euler solvers. The chart below selects one from the model features this study needs. The source tree confirms that `multiphaseEulerFoam` has only drag and heat-transfer interfacial models, with no lift, wall lubrication or turbulent dispersion, which rules it out for pipe bubbly flow.

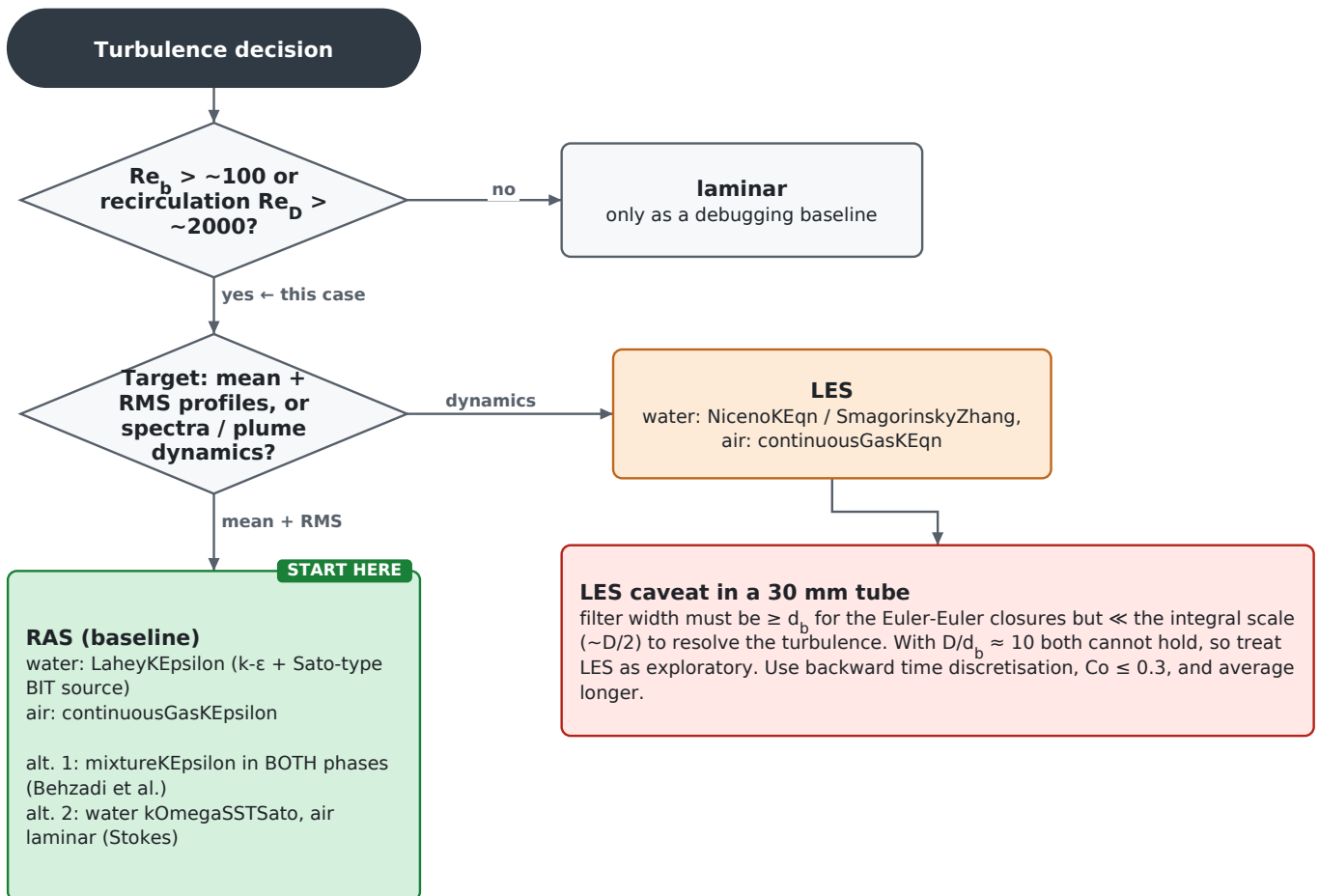


phaseProperties: interfacial closures

Each dispersed-phase model is defined for (air in water). Blending (linear) switches to (water in air) and segregated models in the head space. Surface tension: constant, sigma 0.072. Change **one** closure at a time and record every run.

Force / model	Baseline (v2606 type)	Alternatives in v2606	What it controls, and what to compare
System type	<code>basicTwoPhaseSystem</code>	<code>populationBalanceTwoPhaseSystem</code> (with <code>velocityGroup</code>)	Single size vs. population balance
Bubble diameter	isothermal (d0 from imaging, p0 = sparger p)	constant, <code>velocityGroup</code> (PBE)	Everything. Holdup, slip and lift sign all scale with d_b
Aspect ratio	Wellek	VakhrushevEfremov, Tomiyama, constant	$E = b/a$ vs imaging. Used by TomiyamaAnalytic drag and Lamb VM
Drag	IshiiZuber	TomiyamaCorrelated (A=24 slightly contam., 16 pure), TomiyamaAnalytic, TomiyamaKataokaZunSakaguchi, SchillerNaumann	Slip $U_g - U_l$, gas velocity, global holdup and level swell
Swarm correction	none ($\alpha_g < 5\%$)	Tomiyama ($l = 2$)	Only matters above $\alpha_g \approx 10\%$
Lift	<code>wallDamped { lift Tomiyama; wallDamping cosine Cd 1 }</code>	LegendreMagnaudet, Moraga, constantCoefficient	Radial void profile (wall vs core peak), liquid velocity profile shape
Wall lubrication	Antal ($Cw1 -0.01$, $Cw2 0.05$)	Frank (Cwd , Cwc , p), Tomiyama ($Cwd = D = 0.03$ m)	Void-free layer and position of the near-wall α peak
Turbulent dispersion	LopezDeBertodano ($Ctd 0.25$) for start-up, then Burns ($\sigma 0.9$, $Ctd 1$) from $t \geq 0.2$ s	Gosman, constantCoefficient	Radial spreading of the plume, α profile width. Tested: Burns diverges at $t \approx 0.03$ s from the quiescent column but is stable on restart
Virtual mass	constantCoefficient ($Cvm 0.5$)	Lamb (uses aspect ratio)	Transient response, numerical stability
Heat transfer	RanzMarshall (isothermal anyway)	spherical, constantNu	Irrelevant at 20 °C, but must be present

§3 (cont.) · Turbulence modelling

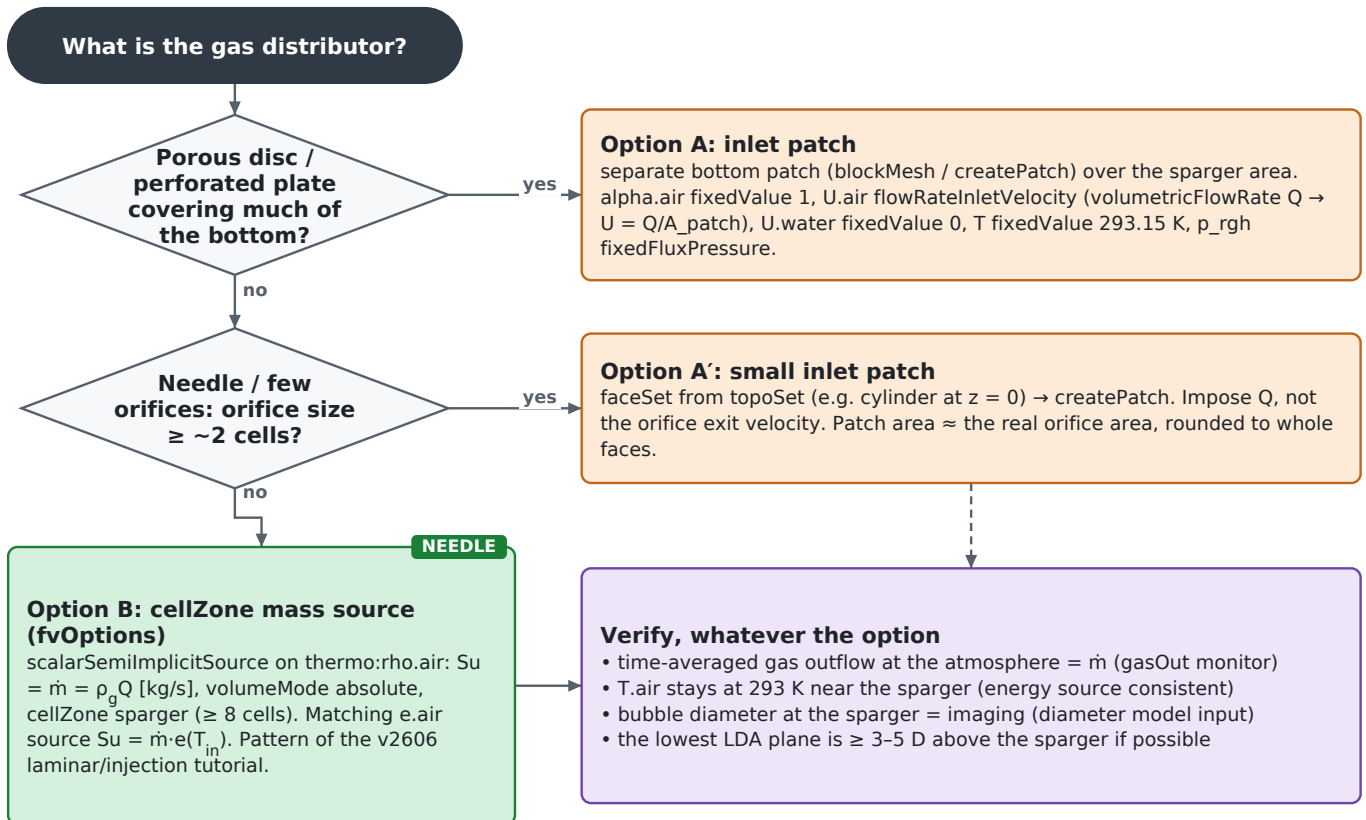


RMS from RAS. RANS gives only the turbulent kinetic energy k , which is isotropic. For the axial RMS compared with LDA use $u'_{z,rms} \approx \sqrt{(\overline{u'_z u'_z})_{resolved} + 2k/3}$, where the resolved part comes from UPrime2Mean of the unsteady RANS. Bubbly pipe flow is strongly anisotropic (axial ≈ 1.5 – $2\times$ radial), so this estimate is expected to under-predict the axial RMS and over-predict the radial one. State this in the comparison.

- The LaheyKEpsilon + continuousGasKEpsilon pair is used together in the v2606 RAS/wallBoiling tutorial, and mixtureKEpsilon (both phases) in RAS/bubbleColumn. mixtureKEpsilon additionally needs the km and epsilon fields in 0/.
- Turbulent dispersion needs turbulence:** Burns and LopezDeBertodano use the liquid ν_t/k , so with a laminar liquid they are zero.
- Keep the fluid properties consistent: the tutorial thermophysicalProperties.water describes hot water ($\rho_0 = 1027$, $\mu = 3.6 \times 10^{-4}$). Set $\rho_0 \approx 998$, $\mu = 1.0 \times 10^{-3}$ Pa·s and $Pr \approx 7$ for 20 °C (template in Appendix A).

§4 · Gas injection

The injection model is the least transferable part of the set-up. It decides how the gas is spread in the first few diameters above the sparger, and Euler-Euler cannot resolve bubble formation. Pick the option from the sparger geometry:



Q [L/min]	$\dot{m} = \rho_g Q$ [kg/s]	e.air source $\dot{m} \cdot e_{in}$ [W]	U at a 12×12 mm patch [m/s]	U at a Ø1 mm orifice [m/s] (do not impose)
0.1	2.000e-06	-0.178	0.0116	2.12
0.2	4.000e-06	-0.357	0.0231	4.24
0.3	6.000e-06	-0.535	0.0347	6.37
0.4	8.000e-06	-0.713	0.0463	8.49
0.5	1.000e-05	-0.892	0.0579	10.61

e_{in} is the sensible internal energy of air at 293.15 K for hConst + perfectGas ($T_{std} = 298.15$ K): $e = C_p(T - T_{std}) - RT = -89169$ J/kg. Without it the injected gas enters with $e = 0$, i.e. at the wrong temperature, and ρ_g drifts.

- Flowmeter units:** rotameters and mass-flow controllers usually read normal or standard L/min (0 °C or 20 °C, 1.013 bar). Convert to actual volumetric flow at the sparger ($p \approx 1.043$ bar, 20 °C) or, better, prescribe the **mass** flow (Option B).
- Start-up:** ramp the source over ~0.5 s (a Function1 table for Su in the source dictionary, or a restart), or begin injecting at $t = 0.1$ s as in the tutorial. This avoids a pressure shock in the still column.
- Momentum source:** not needed for Option B. Bubbles detach with negligible momentum compared with buoyancy. The v2606 injection tutorial shows the vectorSemiImplicitSource pattern if you need one.

§5 · Boundary and initial conditions

Patches: bottom (wall), wall (tube wall), atmosphere (open top, patch), and optionally sparger (Options A/A'). Entries marked ϕ use `phi.air/phi.water` as appropriate.

Field	wall, bottom	atmosphere (top)	sparger (Option A)	Initial value
alpha.air	zeroGradient	inletOutlet inletValue 1 (ϕ)	fixedValue 1	setFields: 0 for $z < 0.30$ m, 1 above
alpha.water	zeroGradient	inletOutlet inletValue 0 (ϕ)	fixedValue 0	1 – alpha.air
U.air	noSlip (or slip, test it)	pressureInletOutletVelocity (ϕ)	flowRateInletVelocity volumetricFlowRate Q	(0 0 0)
U.water	noSlip	pressureInletOutletVelocity (ϕ)	fixedValue (0 0 0)	(0 0 0)
p_rgh	fixedFluxPressure	prghPressure p 1e5 (or prghTotalPressure)	fixedFluxPressure	1e5
p	calculated	calculated	calculated	1e5
T.air, T.water	zeroGradient	inletOutlet 293.15 (ϕ)	fixedValue 293.15	293.15 K
k.*	kqRWallFunction	inletOutlet (ϕ)	fixedValue small	1e-5 m ² /s ²
epsilon.*	epsilonWallFunction	inletOutlet (ϕ)	fixedValue small	1e-5 m ² /s ³
omega.* (SST)	omegaWallFunction	inletOutlet (ϕ)	fixedValue	≈ 1 1/s
nut.*	nutkWallFunction	calculated	calculated	0
alphan.*	compressible::alphanWallFunction Prt 0.85	calculated	calculated	0
km, epsilon (mixing length, Kepsilon)	zeroGradient	inletOutlet phim	fixedValue	as k, epsilon

- **Why this top BC:** prghPressure holds the static pressure at 1 atm. inletOutlet on alpha lets gas leave and allows only air back in, so the head space can breathe as the level swells. The same combination is used in the v2606 bubbleColumn tutorials.
- **Initial state:** quiescent liquid, hydrostatic pressure (the solver builds it from p_rgh and g), no gas in the liquid. setFields with a boxToCell puts the free surface exactly at 300 mm. Stagger the surface on a cell face (with $\Delta z = 2$ mm, 300 mm is a face) to avoid a smeared initial interface.
- **Gas wall BC:** dispersed bubbles do not stick to walls, so slip for U.air is physically reasonable. The v2606 tutorials use no-slip, and near-wall α is small, so the difference is minor. Run it once as a sensitivity check.
- **Residual values:** residualAlpha 1e-6 for both phases (air in the liquid is only $\sim 1\%$), and residualRe 1e-3 in the drag models.

Gravity and reference: constant/g = (0 0 -9.81). If you ever use hRef, set it at the free surface. Otherwise leave it out: $p_rgh = p - \rho g \cdot h$ is referenced to the origin.

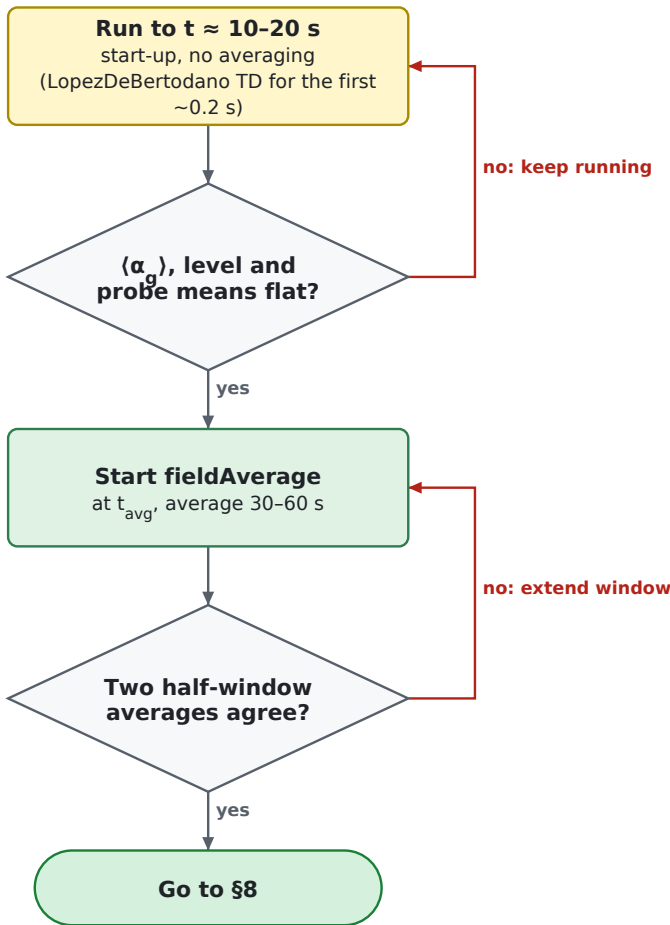
§6 · Numerics and run control

Setting	Recommendation	Reason
Time step	adjustTimeStep yes, maxCo 0.25 (0.2 for LES), maxDeltaT 2e-3 s	Template run (medium mesh, 0.3 L/min): Co-limited $\Delta t \approx 1.1$ ms. Expect smaller Δt at higher Q or on the fine mesh
PIMPLE	nOuterCorrectors 3, nCorrectors 2, faceMomentum yes	faceMomentum suppresses checkerboard velocities at the free surface and near the sparger
Alpha	nAlphaCorr 1, nAlphaSubCycles 3 (MULES)	Keeps α bounded at the interface
ddt	Euler (RANS), backward (LES)	Bounded start-up. Second order for resolved dynamics
div(U)	Gauss limitedLinearV 1	As in the tutorials, bounded yet low diffusion
div(alpha)	Gauss vanLeer	TVD for the phase fraction
div(k, ϵ)	Gauss limitedLinear 1	
Laplacian	Gauss linear corrected	O-grid: non-orthogonality $\leq 30^\circ$
Linear solvers	p_rgh: GAMG/DIC tol 1e-8 relTol 0.01 (Final 0). U, e, k, ϵ : smoothSolver symGaussSeidel 1e-7	
Parallel	decomposePar, scotch, 4–8 cores for ~ 60 k cells	~ 7 –15 k cells per core
Output	writeFormat binary, writeInterval 1 s, purgeWrite as disk allows	Averages live in the function objects, so few time directories are needed

Run-time function objects (system/functions, template in Appendix A)

Function object (type)	Purpose
volFieldValue volAverage alpha.air on cellZone liquidRegion	global gas holdup $\langle \alpha_g \rangle(t)$: steadiness check and comparison with measured holdup/level swell
interfaceHeight on alpha.water	careful: returns $\int \alpha_{\text{water}} dz$ along a vertical line, i.e. the collapsed liquid height (0.284 m on the axis in the template run), not the surface position. Get the swollen level from the $\alpha_{\text{water}} = 0.5$ crossing of a sampled vertical line (sets). $(\text{level} - \text{collapsed height})/\text{level} = \text{line-averaged holdup}$
surfaceFieldValue sum alphaRhoPhi.air on patch atmosphere	gas mass outflow. Its time average must equal $\dot{m}$ (mass balance)
probes at LDA points	U.water, U.air, α , k time series: statistics, convergence, spectra
multiply alpha.air $\times$ U.air $\rightarrow$ alphaU.air	needed for the α -weighted (Favre) gas velocity
fieldAverage (timeStart = t_{avg})	UMean, UPrime2Mean for both phases, alphaMean, alphaU.airMean, kMean
sets (uniform lines)	two orthogonal diameters per LDA height, csv
surfaces (cuttingPlane)	full planes at each LDA height, raw, for azimuthal averaging in Python
yPlus, CourantNo, solverInfo	mesh / time-step / convergence diagnostics
phaseForces (optional)	writes the individual interfacial forces (drag, lift, wall lubrication, TD): useful for diagnosis in §9

§7 · Running and judging statistical steadiness



Monitors to plot during the run

- postProcessing/holdup: $\langle \alpha_g \rangle(t)$ in the liquid region
- postProcessing/freeSurface: collapsed liquid height (t)
- postProcessing/gasOut: gas mass outflow vs $\dot{m}$
- postProcessing/ldaProbes: U.water, U.air at the LDA points

Plot running means of each monitor. Steady state is reached when the running mean of the last ~5 s stops drifting. Fluctuations themselves never stop: the plume keeps meandering.

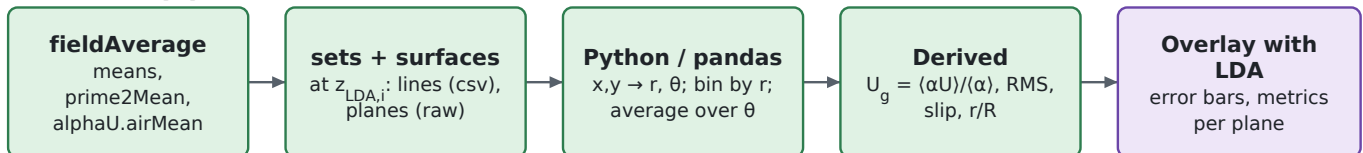
- **Start-up sequence (tested):** with all lateral forces on, the run diverged at $t \approx 0.03$ s, when the first gas left the sparger zone. Switching forces off one at a time showed that Burns turbulent dispersion was the cause; lift and wall lubrication were stable. Start with LopezDeBertodano, then change the turbulentDispersion entry to Burns and restart (startFrom latestTime) at $t \geq 0.2$ s. That restart ran without problems.
- **Averaging length:** at least 5–10 liquid circulation times (~5–6 s each), so 30–60 s. Check convergence by averaging the first and second half of the window separately. The profiles at each LDA plane should agree within ~2–3 % (mean) and ~5–10 % (RMS).
- **Cost (measured):** the medium mesh (59 k cells) ran 3 s of physical time in 29 min on 8 cores ($\Delta t \approx 1.1$ ms). 60 s is therefore ≈ 10 h on 8 cores per flowrate. Do model screening on the coarse mesh (~26 k cells, $\approx 3\times$ faster).
- **Mass balance:** $\dot{m}_{in} = \text{gasOut} + d(\text{gas mass})/dt$. Note that gasOut matches $\dot{m}$ almost from the first time step, because the rising level pushes head-space air out of the top (seen in the template run). So gasOut alone does not show steadiness. The holdup monitor must also have levelled off.

§8 · Post-processing and comparison

LDA gives point-wise, time-series velocities in a small measurement volume ($\sim 0.1 \times 0.1 \times 1$ mm). The simulation gives cell-averaged, phase-averaged fields. Before any plot, map both onto the **same statistics**:

Experimental quantity	How it is measured	Simulation counterpart	Processing / caveats
Mean axial liquid velocity $\langle U_{l,z} \rangle(r)$ at z_i	LDA on seeding particles, radial traverse(s). Validated bursts from liquid only	<code>U.waterMean</code> z-component on sets/planes	Transit-time weighting of LDA samples (removes velocity bias). Azimuthally average the CFD. Mirror or average $\pm r$
Liquid RMS u'_z, u'_r	LDA 2-component, same traverse	$\sqrt{(U.waterPrime2Mean_{zz} + 2/3 k.waterMean)}$	RANS k is isotropic, so expect an under-predicted axial RMS. Report resolved and modelled parts separately
Mean gas (bubble) velocity $\langle U_{g,z} \rangle(r)$	LDA bursts reflected from bubble surfaces (amplitude / burst-length discrimination), or imaging PTV	<code>alphaU.airMean</code> / <code>alpha.airMean</code>	LDA bubble data are bubble-number weighted, which is closer to the α -weighted (Favre) mean than to <code>U.airMean</code> . Ignore cells where α is at the residual level
Slip velocity	$\langle U_g \rangle - \langle U_l \rangle$ from the two LDA channels	same difference from the CFD	Most direct test of the drag model and d_b
Local void fraction $\alpha(r)$	not from LDA. Conductivity/optical probe, or the gas LDA data rate as a proxy (qualitative)	<code>alpha.airMean</code>	If only the data rate exists, compare normalised profiles (shape) and not magnitudes
Global holdup, level swell	free-surface rise from imaging or a ruler, Δp between two taps	holdup monitor. Level from $\alpha_{water} = 0.5$ on sampled vertical lines	Most robust integral check. Do it first. Consistency: $\Delta h \approx H_L \langle \alpha_g \rangle / (1 - \langle \alpha_g \rangle)$
Bubble size d_{32} , distribution	shadowgraphy: projected area $\rightarrow d_{eq}$, $d_{32} = \Sigma d^3 / \Sigma d^2$	<code>d0</code> input (constant/isothermal) or <code>sizeDistribution FO</code> with <code>velocityGroup</code>	An input for the monodisperse model, an output for the PBE. Check the imaging plane against the LDA heights
Aspect ratio $E = b/a$	ellipse fit to bubble contours	E from the Wellet / VakhrushevEfremov model at the measured E_0	Choose the aspect-ratio model that matches, then check its effect on drag
Bubble rise velocity	PTV on images	α -weighted <code>U.air</code> (as above)	Consistency check between the imaging and LDA gas velocities
Spectra, plume wandering frequency	LDA time series (irregular sampling: slotting or resampling)	probes (URANS/LES only)	Only meaningful with a resolved-dynamics approach

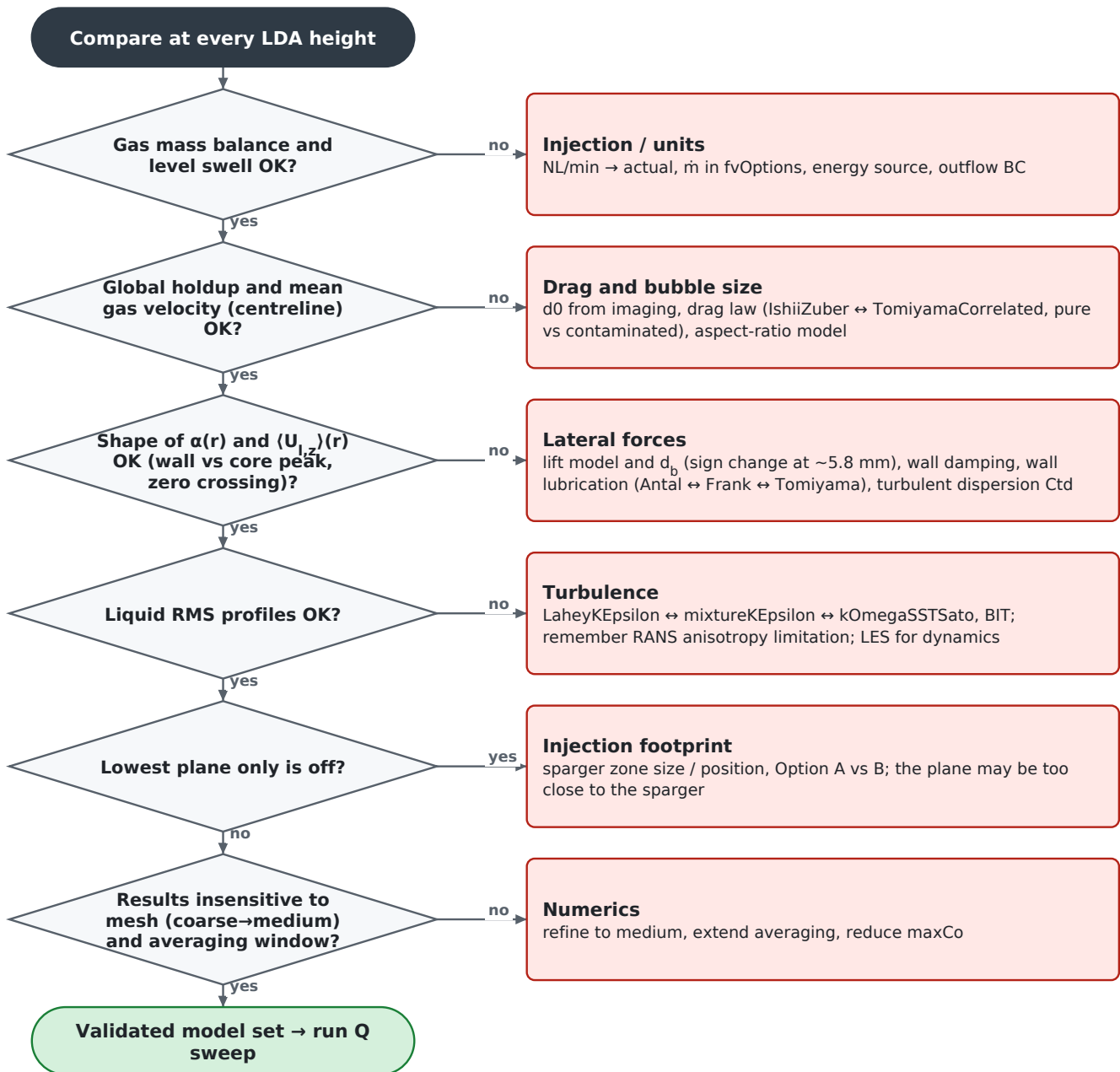
Extraction pipeline



- **Sampling in cells:** use `interpolate false` on the planes (cell values, no smoothing). Bin the radius with $\Delta r \approx$ the local cell size. The O-grid is not polar, so bins at larger r collect more cells, which is fine for averaging.
- **Heights:** choose LDA planes in the developed region, e.g. $z/D \approx 2, 5$ and 8 ($z \approx 60, 150$ and 240 mm). Stay $\geq 1-2 D$ away from the free surface, which oscillates and carries droplets. The template has $z = 50/150/250$ mm. Change `zLDA1..3` in `controlDict`.
- **Re-sampling without rerunning:** `reactingTwoPhaseEulerFoam -postProcess -dict system/functions -latestTime`.
- **Metrics:** for each height, report the normalised RMS error $\|U_{CFD} - U_{LDA}\| / \max|U_{LDA}|$, the centreline value and the zero-crossing radius of $\langle U_{l,z} \rangle$ (upflow/downflow boundary, a sensitive indicator of the lift and TD balance).

§9 · Validation and diagnosis

Work from the most integral quantity to the most local one. Each check points to the closures that drive it. Change **one** model or coefficient per iteration, keep a run log, and do this on the coarse mesh.



Tip: turn on the phaseForces function object for a diagnosis run. It writes the drag, lift, wall-lubrication and turbulent-dispersion force fields, so you can see which force dominates the radial balance at each height before changing a coefficient.

Test matrix and sensitivity plan

Step	Runs	Mesh	Purpose
1 Sanity	Q = 0.3 L/min, baseline models, laminar and RAS	coarse	the set-up works: mass balance, level swell, no divergence
2 Mesh	Q = 0.3, baseline RAS	coarse / medium / fine	plateau of holdup, level swell and centreline U_j ; pick the production mesh
3 Averaging	Q = 0.3, production mesh	medium	t_{avg} and window length (half-window test)
4 Closures	one-at-a-time: drag (3 options) $\times$ lift on/off $\times$ WL (3) $\times$ TD (2) $\times$ turbulence (3)	coarse $\rightarrow$ medium	sensitivity ranking. Keep the best set, not a per-flowrate tuned set
5 Diameter	$d_0 = \text{imaging mean} \pm 1\sigma$; velocityGroup PBE if imaging shows a wide distribution	medium	effect of the size uncertainty (critical near the lift sign change)
6 Production	Q = 0.1, 0.2, 0.3, 0.4, 0.5 L/min with the final model set	medium	comparison with all LDA planes and the imaging data
7 Optional	LES (NicoKEqn / SmagorinskyZhang) at one Q	medium	RMS anisotropy, plume wandering, spectra vs LDA

Information to collect from the experiment before simulating

- Sparger: type, orifice diameter(s), number and layout of holes, height of the needle tip above the bottom.
- Flowmeter: type, calibration gas, reference conditions (normal/standard/actual), accuracy.
- Water: temperature, tap or demineralised (contamination sets the drag regime), any surfactant.
- LDA: measurement heights and radial positions, number of components, how gas and liquid signals are separated, seeding, sample count and sampling time per point, refraction correction, uncertainty estimate.
- Imaging: field of view and heights, resolution (px/mm), bubble detection method, d_{eq} definition, number of bubbles, and whether overlapping bubbles are rejected (this biases d_{32} at higher Q).
- Level swell and/or global holdup for each Q (cheap and very useful).

Key references for the models

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Appendix A · Template case (checked with v2606)

The template was generated by a small Python script (medium mesh, $Q = 0.3$ L/min, Option B). blockMesh, checkMesh (Mesh OK, 58 800 cells), topoSet (sparger zone: 24 cells) and setFields all completed. A parallel run (8 cores) then reached **$t = 3$ s without problems** (29 min wall time, $\Delta t \approx 1.1$ ms). The gas holdup in the liquid levelled off at 3.06 % from $t \approx 1.6$ s (drift-flux estimate in §0: 2.7 %), the free surface ($\alpha_{\text{water}} = 0.5$) rose to $z \approx 309.5$ mm, a level swell of ≈ 9.5 mm, and the gas outflow equalled $\dot{m} = 6.02 \times 10^{-6}$ kg/s. This shows only that the set-up works. 3 s is far too short for any comparison with experiment (§7). The excerpts below are the parts that differ from the stock bubbleColumn tutorial. Coefficients are starting values, not tuned ones.

```
bubblyTube/
├── 0.orig/      alpha.air alpha.water U.air U.water p p_rgh T.air T.water
│               k.* epsilon.* nut.* alphas.* (patches: bottom, wall, atmosphere)
├── constant/   g phaseProperties fvOptions
│               thermophysicalProperties.{air,water} turbulenceProperties.{air,water}
├── system/     blockMeshDict topoSetDict setFieldsDict controlDict functions
│               fvSchemes fvSolution decomposeParDict
└── Allrun      restore0Dir → blockMesh → checkMesh → topoSet → setFields → solver
```

system/blockMeshDict (O-grid, core)

```
R 0.015; s 0.0075; H 0.35;
Rc #eval{ $R*sqrt(0.5) }; sm #eval{ -$s }; Rcm #eval{ -$Rc }; Rm #eval{ -$R };
vertices
(
    ($sm $sm 0) ($s $sm 0) ($s $s 0) ($sm $s 0) // 0-3 core square
    ($Rcm $Rcm 0) ($Rc $Rcm 0) ($Rc $Rc 0) ($Rcm $Rc 0) // 4-7 outer circle
    ($sm $sm $H) ($s $sm $H) ($s $s $H) ($sm $s $H) // 8-11
    ($Rcm $Rcm $H) ($Rc $Rcm $H) ($Rc $Rc $H) ($Rcm $Rc $H) // 12-15
);
Nc 12; Nr 4; Nz 175;
blocks
(
    hex (0 1 2 3 8 9 10 11) ($Nc $Nc $Nz) simpleGrading (1 1 1) // core
    hex (4 5 1 0 12 13 9 8) ($Nc $Nr $Nz) simpleGrading (1 1 1)
    hex (5 6 2 1 13 14 10 9) ($Nc $Nr $Nz) simpleGrading (1 1 1)
    hex (6 7 3 2 14 15 11 10) ($Nc $Nr $Nz) simpleGrading (1 1 1)
    hex (7 4 0 3 15 12 8 11) ($Nc $Nr $Nz) simpleGrading (1 1 1)
);
edges
(
    arc 4 5 (0 $Rm 0) arc 5 6 ($R 0 0) arc 6 7 (0 $R 0) arc 7 4 ($Rm 0 0)
    arc 12 13 (0 $Rm $H) arc 13 14 ($R 0 $H) arc 14 15 (0 $R $H) arc 15 12 ($Rm 0 $H)
);
boundary
(
    bottom { type wall; faces ((0 3 2 1) (4 0 1 5) (5 1 2 6) (6 2 3 7) (7 3 0 4)); }
    wall { type wall; faces ((4 5 13 12) (5 6 14 13) (6 7 15 14) (7 4 12 15)); }
    atmosphere { type patch; faces ((8 9 10 11) (12 8 9 13) (13 9 10 14) (14 10 11 15) (15 11 8 12)); }
);
```

constant/fvOptions (Option B, $Q = 0.3$ L/min)

```
options
{
    airMassSource
    {
        type scalarSemiImplicitSource;
        selectionMode cellZone;
        cellZone sparger; // topoSet: cylinderToCell r = 2.5 mm, z = 0-4 mm
        volumeMode absolute;
        sources { thermo:rho.air (6.0204e-06 0); } // kg/s = rho_g * Q
    }
    airEnergySource
    {
        type scalarSemiImplicitSource;
        selectionMode cellZone;
        cellZone sparger;
        volumeMode absolute;
        sources { e.air (-5.3683e-01 0); } // W = mdot * e(293.15 K)
    }
}
```

constant/phaseProperties (dispersed-phase closures; blending, (water in air) and segregated drag as in the tutorial)

```
type                basicTwoPhaseSystem;
phases              (air water);
air { type purePhaseModel; diameterModel isothermal;
    isothermalCoeffs { d0 3e-3; p0 1.03e5; } residualAlpha 1e-6; } // d0 <- imaging
water { type purePhaseModel; diameterModel constant; constantCoeffs { d 1e-4; } residualAlpha 1e-6; }

aspectRatio ( (air in water) { type Wellek; } (water in air) { type constant; E0 1.0; } );
drag ( (air in water) { type IshiiZuber; residualRe 1e-3; swarmCorrection { type none; } }
    (water in air) { type SchillerNaumann; residualRe 1e-3; swarmCorrection { type none; } }
    (air and water){ type segregated; m 0.5; n 8; swarmCorrection { type none; } } );
virtualMass ( (air in water) { type constantCoefficient; Cvm 0.5; }
    (water in air) { type constantCoefficient; Cvm 0.5; } );
lift ( (air in water) { type wallDamped;
    lift { type Tomiyama; }
    wallDamping { type cosine; Cd 1.0; } } );
wallLubrication ( (air in water) { type Antal; Cw1 -0.01; Cw2 0.05; } );
turbulentDispersion ( (air in water) { type LopezDeBertodano; Ctd 0.25; } ); // start-up
// after t >= 0.2 s (restart): (air in water) { type Burns; sigma 0.9; Ctd 1.0; residualAlpha 1e-4; }
heatTransfer ( (air in water) { type RanzMarshall; residualAlpha 1e-4; }
    (water in air) { type RanzMarshall; residualAlpha 1e-4; } );
phaseTransfer ();
surfaceTension ( (air and water) { type constant; sigma 0.072; } );
pMin 10000;
```

system/functions (excerpt)

```
alphaUair { type multiply; libs (fieldFunctionObjects); fields (alpha.air U.air);
    result alphaU.air; executeControl timeStep; writeControl none; }
average { type fieldAverage; libs (fieldFunctionObjects); timeStart $tAvg; writeControl writeTime;
    fields ( U.water { mean on; prime2Mean on; base time; }
        U.air { mean on; prime2Mean on; base time; }
        alphaU.air { mean on; prime2Mean off; base time; }
        alpha.air { mean on; prime2Mean on; base time; }
        k.water { mean on; prime2Mean off; base time; } ); }
holdup { type volFieldValue; libs (fieldFunctionObjects); fields (alpha.air); operation volAverage;
    regionType cellZone; name liquidRegion; writeFields false; }
freeSurface { type interfaceHeight; // collapsed liquid height, see $6 libs (fieldFunctionObjects); alpha alpha.water;
    locations ((0 0 0) (0.010 0 0)); }
gasOut { type surfaceFieldValue; libs (fieldFunctionObjects); regionType patch; name atmosphere;
    operation sum; fields (alphaRhoPhi.air); writeFields false; }
ldaPlanes { type surfaces; libs (sampling); writeControl writeTime; timeStart $tAvg;
    surfaceFormat raw; interpolate false;
    fields (U.waterMean U.airMean alphaU.airMean alpha.airMean U.waterPrime2Mean k.waterMean);
    surfaces { z1 { type cuttingPlane; planeType pointAndNormal;
        pointAndNormalDict { point (0 0 $zLDA1); normal (0 0 1); } } } }
```